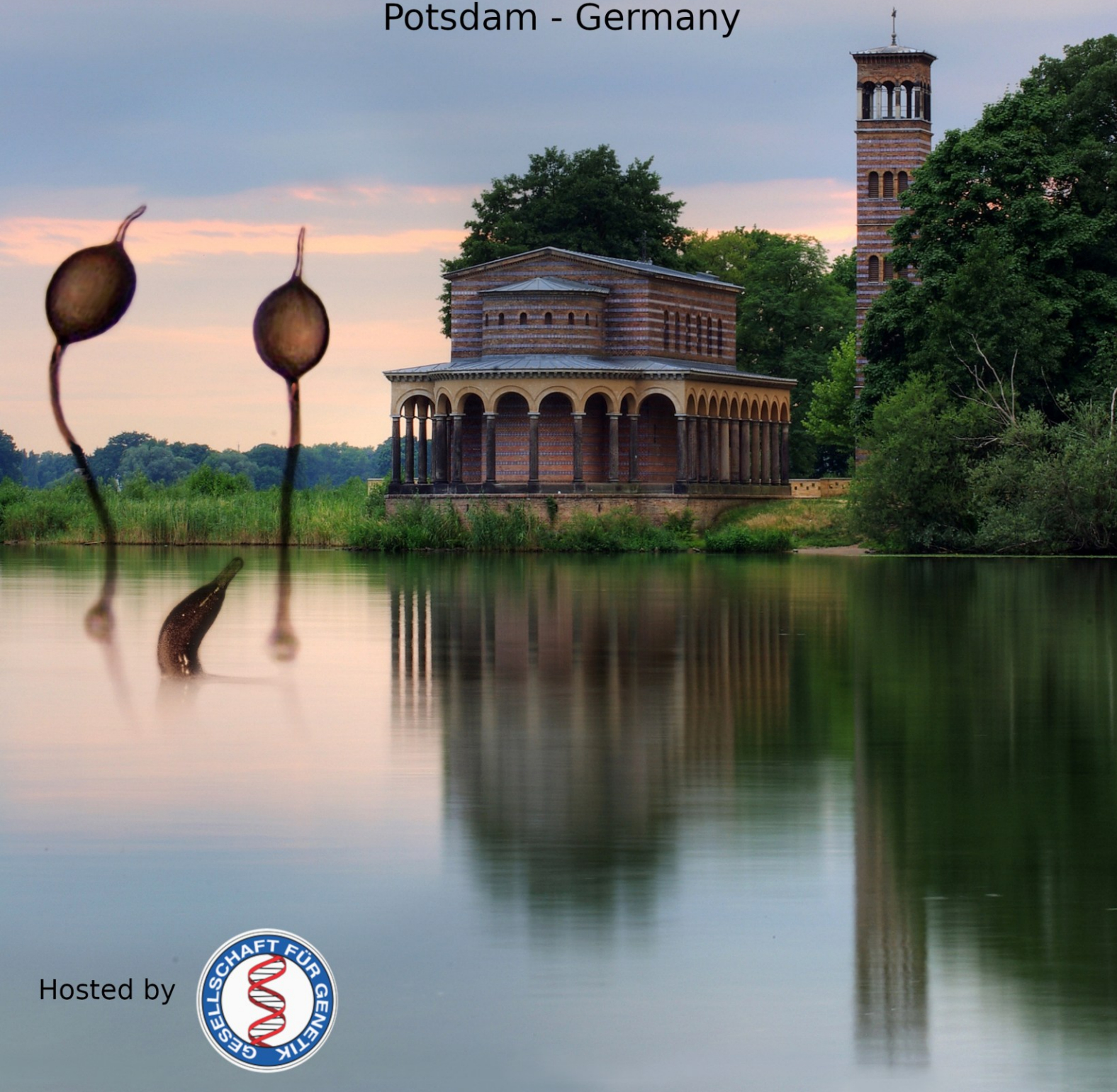


Dicty 2025

Annual International Dictyostelium Conference

August 3rd - August 7th 2025

Potsdam - Germany



Hosted by



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Dicty 2025

August 3 - 7, 2025

Venue:

Kongresshotel Potsdam
Am Luftschiffhafen 1
14471 Potsdam

Organizing committee:

Christian Hammann (Health and Medical University Potsdam)
Ralph Gräf (University of Potsdam)

Dicty 2025 - Program

Sunday, 3.8.25	Monday, 4.8.25	Tuesday, 5.8.25	Wednesday, 6.8.25	Thursday, 7.8.25	
	Bacterial & Pathogen Interactions I 9.00-9.30 Caroline Barisch 9.30-10.00 Pierre Stallforth 10.00-10.30 Daniel S. Stark	Signalling & Chemotaxis I 9.00-9.30 Christopher A. Brimson 9.30-10.00 Xuehua Xu 10.00-10.30 Richard H. Gomer	Cytoskeletal Dynamics 9.00-9.30 Cristina Martinez Torres 9.30-10.00 Joe Tyler 10.00-10.30 Mark Allan C. Jacob	Departure after Breakfast	
	10.30 Coffee break				
	Bacterial & Pathogen Interactions II 11.00-11.30 Thierry Soldati 11.30-12.00 Rosalind J. Allen 12.00-12.30 Ed Culbertson 12.30-12.40 Poster Flash talks	Signalling & Chemotaxis II 11.00-11.30 Alan Kimmel 11.30-12.00 Jonathan Chubb 12.00-12.10 Poster Flash talks 12.10-12.40 Industrial Presentation (Zeiss)	Development 11.00-11.30 Allyson E. Sgro 11.30-12.00 Sarena B. Nagoor Pitchai 12.00-12.30 Heng Liang		
	12.45 Lunch		Packed lunches		
	Disease Models 14.00-14.30 Laura Antón-Esteban 14.30-15.00 Robert J. Huber The Future of Dictybase and the Stock Centre 15.00-15.30 Open discussion	Metabolites I 14.00-14.30 Markus Maniak 14.30-15.00 Kateryna Maslova 15.00-15.30 Johann E. Kufs	13.00-18.00 Excursion to - Marmorpalais - Neuer Garten - Glienicker Brücke (bridge of spies)		
	15.30 Coffee break				
	Keynote Lecture 16.00-17.00 Mark Field Cell Cycle 17.00-17.30 Ian Jones 17.30-18.00 Irene Meyer 18.00-18.10 Poster Flash talks	Metabolites II 16.00-16.30 Robin S. B. Williams Gene Expression/Genomics 16.30-17.00 Fredrik Söderbom 17.00-17.30 Jordi van Gestel 17.30-18.00 Marek Malicki 18.00-18.10 Poster Flash talks 18.10-18.30 Annette Müller-Taubenberger			
	18.00 – 22.00 Welcome Reception Villa Carlshagen	18.30-20.00 Dinner			19.00-23.00 BBQ
		20.00-22.30 Poster Session			

Sunday, August 3

Meeting point: Villa Carlshagen

17:00 Registration
 18:00 Welcome Dinner



Monday, August 4

8:55 Welcome by the organizers

1st morning session: Bacterial and Pathogen Interaction I; Chair: Monica Hagedorn

9:00 A Double-Edged Sword: Divergent Roles of ER-Contact Sites in Mycobacterial Infection

Caroline Barisch

9:30 A chemical radar allows bacteria to detect and kill amoebae

Pierre Stallforth

10:00 The role of endohyphal bacteria in host-Mucorales interactions

Daniel S. Stark

10:30 - 11:00 **Coffee Break**

2nd morning session: Bacterial and Pathogen Interaction II; Chair: Monica Hagedorn

11:00 Give and take, the intricate lipid interface between mycobacteria and their phagocytic host

Thierry Soldati

11:30 Population-density dependent motility and predation behaviour in *Dictyostelium discoideum*

Rosalind J. Allen

12:00 Evolution and function of TIR domains in *Dictyostelium discoideum*

Ed Culbertson

12:30 - 12:40 **Poster Flash Talks**

12:45 - 14:00 **Lunch**

1st afternoon session: Disease Model; Chair: Caroline Barisch

14:00 *Dictyostelium discoideum* as a model to study missense pathogenic mutations of BPAN disease

Laura Antón Esteban

14:30 Examining the cellular role of the enigmatic CLN5 protein in *Dictyostelium discoideum*

Robert J. Huber

The future of Dictybase and the Stock Center

15:00 **Open discussion**

15:30 - 16:00 Coffee break

Keynote Lecture; Chair: Ralph Gräf

16:00 Evolving the cell interior: Adventures in exploiting parasites to inform on eukaryogenesis

Mark C. Field

2nd afternoon session: Cell Cycle; Chair: Alan Kimmel

17:00 Cell cycle heterogeneity encodes developmental timing in *Dictyostelium*

Ian Jones

17:30 Dynamic Mitotic Localization of the Centrosomal Kinases CDK1, Plk, AurK, and Nek2 in *Dictyostelium* and their role in centrosome duplication

Irene Meyer

18:00 **Poster Flash Talks**

18:30 - 20:00 **Dinner**

20:00 - 22:30 **Poster Session I**

Tuesday, August 5

1st morning session: Signalling and Chemotaxis I; Chair: Annette Müller-Taubenberger

9:00 Cells at the precipice: deciding whether to starve or feed

Christopher A. Brimson

9:30 The role of C2GAP1 for gradient sensing

Xuehua Xu

10:00 Two endogenous *D. discoideum* chemorepellents use different mechanisms to induce repulsion

Richard H. Gomer

10:30 - 11:00 **Coffee break**

2nd morning session: Signalling and Chemotaxis II; Chair: Annette Müller-Taubenberger

11:30 Discerning Chemotactic Responses of Bacterial vs. Self-Synthesized Analogs: Biochemical Complexity to Folate/Pteridine Receptor Signaling

Alan Kimmel

11:30 Generation of topological defects for partitioning of cell populations

Jonathan Chubb

12:00 - 12:10 **Poster Flash Talks**

12:10 Industrial Presentation by Zeiss; *Chair: Ralph Gräf*

Hubert Bauch

12:45 - 14:30 **Lunch**

1st afternoon session: Metabolites I; Chair: Pierre Stallforth

14:00 The roles of sterols and sterol esters in growth and development

Markus Maniak

14:30 Investigating the complex regulation of methionine synthesis in *Dictyostelium* and mammalian models

Kateryna Maslova

15:00 Exploring the natural product potential of social amoeba

Johann E. Kufs

15:30 - 16:00 **Coffee break**

2nd afternoon session: Metabolites II; Chair: Pierre Stallforth

16:00 Pharmacological inhibition of ENT1 enhances the impact of specific dietary fats on energy metabolism gene expression

Robin S. B. Williams

2nd afternoon session: Gene Expression/Genomics; Chair: Christian Hammann

16:30 Evolution of microRNAs in Amoebozoa and implications for the origin of multicellularity

Fredrik Söderbom

17:00 Unlocking CRISPR-Cas9 editing for widely diverse Dictyostelid species

Jordi van Gestel

17:30 From Cytoplasm to Nucleus: New Insights into DIRS-1 Replication Cycle

Marek Malicki

18:00 **Poster Flash Talks**

18:10 - 18:30 In memoriam Günther Gerisch

Annette Müller-Taubenberger

18:30 - 20:00 **Dinner**

20:00 - 22:30 **Poster Session II**

Wednesday, August 6

1st morning session: Cytoskeletal Dynamics; Chair: Thierry Soldati

9:00 How intracellular actin waves mediate topological guidance and curvature sensing

Cristina Martinez Torres

9:30 How do you Make a Macropinocytic Cup?

Joe Tyler

10:00 Intersystem Integration through Contractility Kits

Mark Allan C. Jacob

10:30 - 11:00 **Coffee break**

2nd morning session: Development; Chair: Ludwig Eichinger

11:00 Multicellular development in complex environments

Allyson Sgro

11:30 Impact of simulated heatwaves on multicellularity in *Dictyostelium discoideum*: A proteomics perspective

Sarena Banu Nagoor Pitchai

12:00 *D. discoideum* amoebae fed on poor prey proliferate less but do not lose in the competition to become spores rather than stalk cells

Heng Liang

12:30 Concluding remarks and Poster Prize

12:35 **Pick up packed lunches**

13:00 - 18:00 *Excursion by Bus to*

- **Marmorpalais**
- **Neuer Garten**
- **Glienicker Brücke (Bridge of Spies)**

19:00 **BBQ** (at the hotel)

Thursday, August 7

Breakfast and Departure

Talks

in order of appearance in the program

Bacterial and Pathogen Interactions

A Double-Edged Sword: Divergent Roles of ER-Contact Sites in Mycobacterial Infection

Anna-Carina Mazur¹, Aby Anand¹, Francois Letourner², Patricia Rosell-Arevalo³, Rico Franzkoch⁴, Leonhard Breitsprecher⁴, Stevanus Listian⁵, Iris Hube⁵, Sylvana V. Hüttel¹, Simone Vormittag⁶, Hubert Hilbi⁶, Markus Maniak⁷, Maximiliano G. Gutierrez³, Caroline Barisch¹

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2. Laboratory of Pathogen–Host Interactions, Université de Montpellier, CNRS, INSERM
3. Host–Pathogen Interactions in Tuberculosis Laboratory, The Francis Crick Institute, London, United Kingdom
4. Integrated Bioimaging Facility, Center of Cellular Nanoanalytics, University of Osnabrück, Osnabrück, Germany
5. Division of Molecular Infection Biology, Department of Biology & Center of Cellular Nanoanalytics, University of Osnabrück, Osnabrück, Germany
6. Institute of Medical Microbiology, University of Zürich, Zürich, Switzerland
7. Department of Cell Biology, University of Kassel, Kassel, Germany

Several intracellular pathogens, including *Mycobacterium tuberculosis* (Mtb), disrupt endomembranes to impair immune functions and promote cytosolic escape. In response, host cells initiate membrane repair mechanisms to contain the pathogen. A recently described ER-dependent repair pathway, independent of ESCRT and sphingomyelin, enables damaged lysosomes to form contact sites with the ER, allowing lipid transfer for membrane restoration.

We provide evidence that ER-dependent repair plays a role during mycobacterial infection. Transcriptomic analyses of Mtb-infected macrophages and *M. marinum*-infected *Dictyostelium discoideum* show upregulation of ER-contact site proteins at late infection stages, when vacuolar damage and cytosolic translocation typically occur. In line with this, OSBP in humans and its *D. discoideum* homologue OSBP8 are recruited to MCVs in an ESX1-dependent manner. OSBP8 knockout causes PI4P retention on damaged lysosomes and MCVs, impairs vacuolar function, and reduces cell viability upon sterile damage. These findings suggest OSBP8 contributes to ER-dependent repair by maintaining PI4P homeostasis.

Strikingly, we also observe ER-contact formation with cytosolic mycobacteria, suggesting additional roles of the ER beyond vacuolar maintenance. We hypothesise that VAP, an ER-resident protein anchoring FFAT-motif-containing lipid transporters at membrane-contact sites, is critical in this context. In VAP knockout cells, bacterial growth is attenuated, with an even stronger effect seen in cells expressing FFAT-binding-deficient VAP mutants. This points to a role for VAP in host-pathogen interactions that might extend beyond membrane repair.

A possible explanation is that loss of VAP impairs xenophagy by disrupting autophagosome formation or restricts lipid/nutrient supply to cytosolic mycobacteria. Thus, while OSBP8 appears to function primarily in ER-dependent membrane repair, VAP likely supports additional ER-host interactions that modulate the fate of cytosolic bacteria.

A chemical radar allows bacteria to detect and kill amoebae

Pierre Stallforth, Shuaibing Zhang

Leibniz Institute for Natural Product Research and Infection Biology Leibniz-HKI and Friedrich Schiller University Jena

Amoebal predation drives bacterial evolution, yet little is known about how bacteria detect and counter microbial predators. We show that *Pseudomonas syringae* senses and kills the social amoeba *Polysphondylium pallidum* using a chemical radar system. The bacterium secretes syringafactin, which the amoeba modifies into a signal sensed by the CraR protein, triggering production of the amoebicide pyrofactin. This defense system is widespread in *P. syringae* and enhances plant infection in the presence of amoebae.

Reference: S. Zhang, K. Schlabach, V. H. Perez Carillo, A. Ibrahim, S. Nayem, A. Komor, R. Mukherji, S. Chowdhury, L. Reimer, F. Trottman, A. C. Vlot, C. Hertweck, U. A. Hellmich, P. Stallforth* *Cell* 2025, 188, 2495.

The role of endohyphal bacteria in host-Mucorales interactions

Daniel S. Stark¹, Dora E. Corzo-Leon², Jack Gregory², Andreas Guerrero³, Jessie Uehling³, Elizabeth R. Ballou², Jason S. King¹

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2. MRC Centre for Medical Mycology, University of Exeter, Geoffrey Pope Building, Stocker Road, Exeter, EX4 4QD, UK
3. Department of Botany and Plant Pathology, Oregon State University, Cordley Hall 4625, 2701 SW Campus Way, Corvallis, OR 97331, USA

The Mucorales are an order of soil-based fungi that cause mucormycosis, a disease with few effective treatments and high morbidity and mortality rates in immunocompromised patients. Interestingly, Mucorales have been found to harbour endohyphal bacteria, which can modulate fungal metabolism and provide defence against fungivorous phagocytes. We hypothesise that fungal acquisition of endohyphal bacteria was triggered by selection pressure from environmental predators, but in the process has also provided Mucorales with defence against human phagocytes. The presence of the bacterial endosymbiont *Ralstonia pickettii* in the fungus *Rhizopus microsporus* reduces growth and phagocytosis in *Dictyostelium discoideum*, a social amoeba and soil predator (Itabangi et al., 2022). The presence of endohyphal *R. pickettii* also protects *R. microsporus* spores from clearance by human macrophages and increases fungal virulence in both zebrafish and mouse infection models. We aim to determine the extent and diversity of endohyphal bacteria in other Mucorales species and have screened conditioned media from 60 clinical isolates for *D. discoideum* growth inhibition. We have found that the fungus rather than the bacteria is the main factor affecting growth inhibition, although other *D. discoideum* cellular processes such as phagocytosis, macropinocytosis and bacterial killing are inconsistently impacted between isolates, suggesting that there are diverse mechanisms. We aim to further characterise the mechanisms of anti-phagocyte activity and to determine whether Mucorales-induced *Dictyostelium* phenotypes correlate with virulence in a zebrafish model.

Give and take, the intricate lipid interface between mycobacteria and their phagocytic host

Thierry Soldati

Department of Biochemistry, Faculty of Science, University of Geneva, 30 quai Ernest Ansermet, Sciences II, CH-1211-Genève-4, Switzerland

Intracellular mycobacteria share a conserved infection cycle characterized by damage to their mycobacterium-containing vacuole (MCV) mediated by proteinaceous effectors and lipidic virulence factors. The host reacts to damage by subsequently engaging the ESCRT and autophagy membrane repair pathways. Using both mutants of the bacteria and of the host amoebae and macrophages, we decipher the complex and conserved mechanisms of virulence and cell-autonomous immunity. Membrane microdomains enriched in sterols and stabilized by flotillins are essential for the partitioning of EsxA in the MCV membrane, which is facilitated by PDIMs. Interference with microdomain components dramatically increase resistance to infection. During these stages, components from the mycobacteria cell wall such as PGL are shed and traffic from the MCV to host endosomal compartments and are finally secreted to reach bystander cells. Using selective monitoring of capsular components, we trace their fate and recognition by cytosolic lectins. Interestingly, knockout of lectins strongly restrict intracellular growth. Finally, during infection mycobacteria rely on host-derived lipids as carbon sources, but the dynamics of lipid acquisition remains underexplored, especially in relation to the dual compartmentalization of mycobacteria. When contained inside the MCV and later when released into the cytosol, mycobacteria can access various lipid sources, such as membrane phospholipids and lipid droplet components. Recent results indicate that interference with sterol homeostasis induced by knockout of the NPC sterol transporters in amoebae and macrophages severely impact mycobacteria intracellular growth. Our integrated investigations of the lipid interface so crucial to mycobacteria infection are shedding light on conserved and essential mechanisms.

Population-density dependent motility and predation behaviour in *Dictyostelium discoideum*

Joel Ching Kuma Mbanghanh^{1,2,3}, Silvia Vareschi^{1,2,3}, Marco Mauri^{1,2}, Aishwarya Bhosale^{1,2}, Ariane Zander^{1,2}, Tatjana Malycheva^{2,5}, Lukas Korn^{2,5}, Suman Chakraborty^{2,5}, Nia Verdon^{1,2}, Rosa Herbst^{2,4}, Stefan Schuster^{2,5}, Pierre Stallforth^{2,3,4}, Jonathan Chubb⁶, Rosalind J. Allen^{1,2}

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3. Jena School for Microbial Communication, Jena, Germany
4. Leibniz Institute for Natural Product Research and Infection Biology, Jena, Germany
5. Matthias-Schleiden-Institut, Friedrich Schiller University Jena, Germany
6. UCL Laboratory for Molecular Cell Biology, University College London, United Kingdom

Dictyostelium discoideum (Dicty) cells are well-known to show density-dependent aggregation under starvation conditions, but less is known about whether the behaviour of Dicty cells is also density-dependent under non-aggregating conditions. Using microscopy setups in liquid and on agar, we track the motion of individual Dicty cells, under conditions of high and low population density. Characterising the trajectories followed by the Dicty cells, we find that Dicty motion is population-density dependent. Under conditions of low population density, Dicty cells are more likely to follow extended trajectories, while under conditions of high population density, they are more likely to follow compact trajectories. We also implement individual-based computer simulations that can recapitulate some of this behaviour. Extending our study to investigate the predation behaviour of Dicty, we also characterize the motion of individual Dicty cells on agar surfaces close to colonies of edible and toxic bacteria. We find no evidence that Dicty can sense toxic bacterial colonies. Rather, most Dicty cells migrate towards a bacterial colony, regardless of whether it is toxic or not; if the colony is toxic, the amoeba tend to be killed. However we observe a sub-population of Dicty cells that does not migrate towards a bacterial colony (whether toxic or not). The prevalence of these “non-responder” cells depends strongly on the population density of the Dicty. We speculate that the non-responder cells might form a bet-hedging strategy, allowing the population to avoid extinction in the presence of toxic bacteria.

Evolution and function of TIR domains in *Dictyostelium discoideum*

Ed Culbertson, Emily Cruz-Lorenzo, Neha Garlapati, Kai Fluet, James Drurey, Tera Levin

University of Pittsburgh, Department of Biological Sciences

All cells require cell autonomous innate immunity to both recognize and respond to threats. Innate immunity is not only required for defense, but for many microeukaryotes, like amoeba, it can be used to gather nutrients. The professional phagocyte and social amoeba *Dictyostelium discoideum* (*Dicty*) has been well documented as a voracious bacteriovore. However, little is known about the immune proteins of *Dicty*. What are the immune components of *Dicty*? Are they shared with other microeukaryotes? One protein, TirA, has been shown to be critical for bacterial grazing, as well as both generation of reaction oxygen species, and DNA trap production. TirA is a multidomain protein, with a Toll-interleukin-1 receptor (TIR) domain. TIR domains are innate immune modules found across the tree of life and generally function as structural adaptors, such as TIR proteins in animals, or active NADase enzymes, as in bacteria. In addition to TirA, *Dictyostelium* genomes have two other TIR domain-containing proteins, TirB and TirC, whose roles have not yet been discovered.

We asked the question; are there other TIR domain-containing proteins in *Dicty*, and if so, what might they be doing? We used deep homology to search for TIR domains in amoeba genomes and across the eukaryotic tree of life. We found two additional TIR domain-containing proteins in *Dictyostelium*, which we have named TirD and TirE. TirD is similar to TirB and C, while TirE is highly diverged and shares some of its domain architecture with TirA. We find that the amoeba TirB, C, and D family arose from a recent horizontal gene transfer (HGT) event from bacteria into eukaryotes. In contrast, TirA and E are much more ancient, similar to other TIR-containing proteins across other amoebae. To test if these TIR genes are used for immunity, we generated mutants of TirB, TirC, and TirE in *D. discoideum*. Here, we present phenotypic data on the role of these proteins in amoeba growth, their ability to kill and engulf bacteria as prey, and their susceptibility to infection by the bacterial pathogen *Legionella*. Overall, this work seeks to address a gap in our understanding of the immune systems of diverse eukaryotes.

***Dictyostelium discoideum* as a model to study missense pathogenic mutations of BPAN disease**

Laura Antón-Esteban, Laura Collado, Lucía Lorenzo-Parodi, Olivier Vincent, Ricardo Escalante

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Autophagy is an intracellular degradation process of damaged or unnecessary cytoplasmic components for their destruction or recycling. Since the molecular machinery of autophagy is well conserved between *Dictyostelium discoideum* and humans, it constitutes an excellent model to study the molecular mechanisms of autophagy, and therefore, autophagy-related human diseases. One of them is the beta-propeller protein-associated neurodegeneration (BPAN), a rare disease caused by mutations in the human gene encoding the WIPI4 protein. This protein is a member of the conserved WIPI family with a characteristic 7-bladed beta-propeller structure, and plays a crucial role in the autophagy process, but the molecular and cellular mechanisms of the disease are not fully understood.

Previously, we generated a *Dictyostelium* model of BPAN by disrupting Wdr45l, the WIPI4 homologue, and used it to explore the influence of BPAN missense mutations on the protein function. These studies revealed the presence of two functionally distinct domains that affect the protein's function and subcellular localization (1). For this study, we have introduced three new patient missense mutations in the conserved residues of Wdr45l: V68E, L153R and G161E. We have measured the impact of the mutations on the localization and function of the protein analysing the protein subcellular localization, *Dictyostelium* development, protein levels and autophagic flux. The V68E mutation caused a complete impairment of autophagy, while the L153R and G161E mutations have a milder impact and appeared to affect protein stability. These results show how different pathogenic mutations alter different aspects of the Wdr45l function, and further validate our model of the disease and its use in the analysis of variants of uncertain significance.

1. Tornero-Écija A, Tábara LC, Bueno-Arribas M, Antón-Esteban L, Navarro-Gómez C, Sánchez I, Vincent O, Escalante R. A *Dictyostelium* model for BPAN disease reveals a functional relationship between the WDR45/WIPI4 homolog Wdr45l and Vmp1 in the regulation of autophagy-associated PtdIns3P and ER stress. *Autophagy*. 2022 Mar;18(3):661-677. doi: 10.1080/15548627.2021.1953262. Epub 2021 Jul 27.

Examining the cellular role of the enigmatic CLN5 protein in *Dictyostelium discoideum*

Samer Owiar¹, William D. Kim¹, Robert J. Huber²

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CLN5 is a lysosomal enzyme that participates in a variety of cellular processes such as autophagy and protein transport. Mutations in *CLN5* causes a severe form of neurodegeneration called CLN5 disease, which is a subtype of neuronal ceroid lipofuscinosis (Batten disease). However, the role of the protein in the onset and progression of the disease remains unclear and the precise function of CLN5 is debated. Previous work in *D. discoideum* revealed that CLN5 has glycoside hydrolase activity, with subsequent studies suggesting that the protein functions as a depalmitoylase or bis(monoacylglycero)phosphate (BMP) synthase. To further study the cellular role of CLN5, we used acyl-resin assisted capture (acyl-RAC) to examine the potential depalmitoylase activity of Cln5 in *D. discoideum*. The catalytic triad that regulates the depalmitoylase activity of human CLN5 (Cys²⁸⁰-His¹⁶⁶-Glu¹⁸³) is conserved in *D. discoideum* Cln5. In addition, we previously revealed the Cln5 interactome in *D. discoideum*. This knowledge, combined with the online prediction software SWISS-PALM, allowed us to identify Cln5-interactors that were potentially palmitoylated in *D. discoideum*. Here, we show that acyl-RAC can be used to isolate palmitoylated proteins from *D. discoideum* and that loss of *cln5* prevents the depalmitoylation of three Cln5-interactors. Overall, our findings provide new insight into the function of Cln5 in *D. discoideum* and identify a new tool for studying palmitoylation in this model organism.

Evolving the cell interior: Adventures in exploiting parasites to inform on eukaryogenesis

Mark C. Field

School of Life Sciences, University of Dundee, Dundee, Scotland and Biology Centre, Czech Academy of Sciences, Institute of Parasitology, České Budějovice, Czech Republic.

The origin of eukaryotes represents one of the major evolutionary transitions. It is now widely accepted that the eukaryotic lineage arose within the Archaea, most likely from the Asgard group, which are the closest known prokaryotic relatives of eukaryotes. The defining features of eukaryotic cells include complex internal architectures primarily composed of lipid membrane-enclosed structures, such as the nucleus, Golgi complex, endosomes, and endoplasmic reticulum (ER), each characterized by distinct lipid and protein compositions. In this presentation I will explore contributions toward understanding the fundamental complexity of eukaryotic cells and how this complexity likely emerged through the expansion and contraction of small GTPase families, which likely evolved to provide a specificity system for organelle identity. Additionally, I will examine the origins and modern forms of the protocoatome protein family, which constitutes a widely distributed eukaryotic membrane-deforming system, as well as the ancient multifunctional ESCRT complex, which has taken a distinct pathway. Finally, I will highlight studies in trypanosomatids—an early-branching eukaryotic lineage—focusing particularly on the composition and function of the nuclear pore complex and the nuclear lamina. Collectively, these lines of evidence underscore the amalgamation of both broadly conserved and novel molecular systems present in extant organisms.

Cell cycle heterogeneity encodes developmental timing in *Dictyostelium*

Ian Jones, Jonathan Chubb

University College London

Across multiple biological contexts, there is a close relationship between cell cycle regulation and fate determination. In *Dictyostelium*, the cell cycle position at starvation (the trigger for development) is known to bias fate; with G2 promoting the spore fate, and S/M, the stalk fate. However, the mechanisms governing this process remain elusive.

Through integration of four independent scRNA datasets, we observed a clear early marker of the pre-spore fate, in the enrichment of mitochondrial and cell cycle gene expression. Imaging studies showed that this was reflected in an almost two-fold increase in mitochondrial volume fraction (MVF) between the pre-spore and stalk fates in late development at the 'slug' stage. This was driven by a large mitochondrial synthesis step; immediately following cell division, unique to pre-spore cells and approximately 8-12 hours into development. Cells arrested at S-phase prior to development, known to enrich for the stalk fate, exhibited a 2hr delay in cell division and mitochondrial synthesis. G2 arrested cells showed the converse, dividing and synthesizing mitochondria 2hrs earlier. PspA expression timing exhibited a similar sensitivity to cell cycle position indicating earlier adoption of the spore fate in G2 cells. Theoretical modelling revealed that delayed adoption of the spore fate/ entry into development in S/M-phase cells is sufficient to explain their stalk bias, even in M/S synchronized populations.

Cell cycle heterogeneity in the pre-starvation population was carried forward through development, encoding the timing of mitochondrial synthesis and spore fate acquisition. The cell cycle may act as a general source of heterogeneity for fate decisions in proliferating cell populations.

Dynamic Mitotic Localization of the Centrosomal Kinases CDK1, Plk, AurK, and Nek2 in *Dictyostelium* and their role in centrosome duplication

Stefan Krüger, Nathalie Pfaff, Ralph Gräf, and Irene Meyer

Department of Cell Biology, University of Potsdam, Karl-Liebknecht-Str. 24-25, 14476 Potsdam-Golm, Germany

Dictyostelium exhibits an unusual centrosome cycle, in which duplication is initiated only at the G2/M transition and occurs entirely during the M phase. Little is known about the role of centrosomal kinases in this process. Therefore, we have generated knock-in strains for Aurora (AurK), CDK1, cyclin B, Nek2, and Plk, in which the endogenous kinases are replaced by green fluorescent Neon fusion proteins, and expressed via the endogenous promoters. Our live cell imaging studies show that CDK1 and cyclin B arrive at the centrosome first, already during G2, followed by Plk, Nek2. Interestingly, centrosomal localization patterns of cyclin B and CDK1 were not identical showing that both proteins associate with centrosomes not necessarily as a joint complex. AurK arrives at the centrosome only after centrosome splitting in early mitosis. Thus it plays no role in early centrosome duplication. Furthermore, CDK1/cyclin B and AurK were dynamically localized at kinetochores, and AurK in addition at nucleoli. Surprisingly AurK displayed an uneven distribution between the two daughter cells. With regard to regulation of centrosome duplication we focused on the dynamics of CP39 at the central layer of the centrosomal core structure, where the protein usually disappears at the G2/M transition and reappears in late mitosis. Therefore we created non-phosphorylatable or phosphomimetic mutations at putative phosphorylation sites the respective GFP-CP39 fusion proteins. This showed that mutations at the amino acid positions S184 and S57 of CP39 had an influence on the behavior of the protein. Moreover, yeast-two-hybrid (Y2H) and bimolecular fluorescence complementation experiments revealed a direct interaction between CP39 and CDK1, and Y2H additionally revealed an interaction of cyclin B with Cep192, the mayor component of the outer layers of the centrosomal core structure. Taken together our results support a major role of CDK1/cyclin B in regulation of centrosome duplication.

Cells at the precipice: deciding whether to starve or feed

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From single amoebae to complex multicellular collectives, cells are continually bombarded by many different signals. How can they react appropriately to this complexity of inputs? This question has remained challenging to address, largely because standard reductionist approaches push cell decision making into non-physiological regimes. We now approach this question using the *Dictyostelium* feeding front, where cells have the option to continue migrating towards bacteria and feeding, or to remain behind and enter the developmental programme. I will describe how we are implementing different sensors of extracellular signals and cell state to document the signalling history of individual cells, within the cell collective, to define the rules of signal integration underlying cell choices.

The role of C2GAP1 for gradient sensing

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How do eukaryotic cells translate a steady or a changing chemoattractant gradient into intracellular responses to achieve directional cell migration remain largely elusive. We have previously shown that local recruitment of C2GAP1 controls G protein coupled receptor mediated adaptation and chemotaxis in *D discoideum*. Here, we quantitatively characterized the dynamic of signaling components for gradient sensing in both wild-type or *c2gapA*⁻ cells in response to a steady and/or a changing gradient. Our data reveals the essential role of C2GAP1 in mediating the gradient sensing in response to a steady or changing gradient for proper chemotaxis.

Two endogenous *D. discoideum* chemorepellents use different mechanisms to induce repulsion

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The directed movement of eukaryotic cells is critical for processes such as development and immune responses. While much is known about chemoattraction, much less is known about chemorepulsion. *Dictyostelium discoideum* secretes a 60 kDa chemorepellent protein called AprA to cause cells at the edge of a colony to move away from the colony. In addition to AprA, cells secrete a < 10 kDa chemorepellent. We found that the < 10 kDa chemorepellent is a polymer of phosphates (polyphosphate; polyP). *D. discoideum* cells move by activating cortical Ras at one edge of the cell to initiate pseudopod formation. AprA induces repulsion by inhibiting Ras activation and pseudopod formation on the side of the cell closest to the source of AprA, without affecting the overall frequency of pseudopod formation. In contrast, polyP activates Ras at multiple regions of the cortex, and increases pseudopod formation frequency, especially at the side of the cell furthest from the source of polyP. At least 20 signal transduction proteins are needed for both AprA and polyP repulsion, 9 are needed by polyP but not AprA, and 4 are needed by AprA but not polyP. Proteomics indicated that AprA significantly affects levels of at least 196 proteins and phosphorylation of 139 proteins, and polyP affect levels of at least 2 proteins and phosphorylation of 24 proteins. Except for 11 proteins whose phosphorylation was decreased by both AprA and polyP, these are different sets of proteins. Together, these results indicate that proliferating *D. discoideum* cells use two different chemorepellents, that one of the repellents is the unusual molecule polyphosphate, and that the two repellents activate partially overlapping and partially different pathways to induce repulsion by two basically different mechanisms.

Discerning Chemotactic Responses of Bacterial vs. Self-Synthesized Analogs: Biochemical Complexity to Folate/Pteridine Receptor Signaling

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The recognition and killing of infectious bacteria by *Dictyostelium*, macrophage, and other phagocytic cells is a paramount defense strategy within the innate immune response. Multiple compounds are released by bacteria, and different phagocytic cells express specific chemoattractant receptors with distinct binding specificities; activation of these receptors directs a complex series of intracellular signaling pathways that guide directed cell migration toward the bacterial source. However, many bacterial chemoattractants are structurally related to molecules that are synthesized by and, thus, intrinsic to phagocytic cells. We discuss several modes used by *Dictyostelium* to discriminate bacterial from endogenous Folate analogs. *Dictyostelium* express several folate/pteridine GPCRs; they exhibit distinct affinity response strengths to different Folate analogs and polarize toward different downstream activation pathways.

Generation of topological defects for partitioning of cell populations

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The development of multicellular structures requires spatial partitioning of cells into functional groups. In some embryonic contexts, signal diffusion or local cell interactions are used to make groups of cells distinct. Over larger tissue scales, where diffusion is limiting, additional mechanisms are required to partition cell populations. Here, we demonstrate how *Dictyostelium* regulate the spatial distribution of cAMP signalling territories through the dynamics of topological defects in cell signalling states. By tracking the birth and fate of topological defects and analysing their population dynamics, we show that defects rapidly emerge as cells initiate signalling. As signalling synchronises, topological defects are pruned to a distribution that sets the final territory sizes. A window of competence for stabilisation of territories was revealed by optogenetic culling or ectopic initiation of defects. Modelling implies the robust establishment of defect patterns requires heterogeneity in cell responsiveness to signalling, with some defects emerging as cAMP waves collide with non-responding cells. The model predicts the scaling of territory size with cell density due to differences in wave dynamics at different densities. These predictions were validated by measurements of natural waves, and perturbation of wave dynamics. Overall, our data reveal partitioning is governed by cell heterogeneity driving signalling centre formation, and wave dynamics determining signalling centre stability.

The roles of sterols and sterol esters in growth and development

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Over 50 years ago, when Long and Coe followed lipid profiles of *Dictyostelium* cells throughout development, they described a „substance G“ that was barely detectable in vegetative amoebae, steadily increased from aggregation phase to the mature fruiting body, and then began to decline slowly. It was suggested that this substance comprised steryl-esters.

We have repeated and confirmed their observations and set out to investigate the biological role of steryl-esters and their main constituent, sterol. Lipase 5 is an enzyme detected on biochemically prepared lipid droplets, and its overexpression under control of a *csA* promotor during development reduces the levels of steryl-esters markedly. While the fruiting body develops normally, a fraction of spores is stained by the vital-dye trypan blue and also fails to germinate on a bacterial lawn. Knocking out the two candidates for sterol-ester synthases fully abolishes the production of steryl-esters in development leading to a much stronger decline in spore viability and germination. A somewhat intermediate effect on the activity of spores is observed upon deletion of steryl-methyl-transferase B (*SmtB*). In conclusion, steryl-esters are needed to maintain spores with the ability to germinate over a week or more after their formation.

Because *SmtB* knockout cells should also affect the structure of the free sterol molecule, we also looked at a possible impact on vegetative cells and found a prolonged doubling time in axenic culture. Terbinafine, a well-known inhibitor of sterol synthesis, fully blocks cell division without killing the cells. Together, these observations indicate that steryl-esters are stored in spores to provide the amoeba with essential sterols during or immediately after germination.

Investigating the complex regulation of methionine synthesis in *Dictyostelium* and mammalian models

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The folate one – carbon cycle (FOCC) is a complex cellular pathway leading to the synthesis of the essential amino acid, methionine, the precursor of the universal methyl donor, S-adenosylmethionine (SAM), used in epigenetic modification in health and disease. However, our understanding of the regulation of methionine synthesis is limited. In this project, we employ *Dictyostelium* as a model to investigate the regulation of methionine synthesis, focusing on the enzymes responsible for methionine production: cobalamin-dependent methionine synthase (MTR) and cobalamin-independent methionine synthase (*metE*). We demonstrate that the *D. discoideum* and *H. sapiens* MTR enzymes are highly conserved at the amino acid level and in tertiary structure. Individual ablation of both genes in *mtr⁻* and *metE⁻* cells enabled the analysis of each pathway independently to show compensatory transcriptional mechanisms. Use of a pharmacological inhibitor of MTR identified a reduction in methionine levels in *metE⁻* cells and the regulation of other key FOCC metabolites. Finally, we extend our findings to a mammalian neuronal cell line model to show similar regulation of FOCC metabolites using pharmacological inhibitors, and we identify changes in FOCC-related proteins through proteomics analysis and western blot validation. Overall, our research provides novel insights into methionine regulation through targeting pathways of methionine synthesis, suggesting potential therapeutic approaches to modulate downstream metabolic and epigenetic effects.

Exploring the natural product potential of social amoeba

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In addition to its remarkable life cycle, the social amoeba *Dictyostelium discoideum* is of particular interest for natural product research due to the wealth of biosynthetic genes encoded in the genome. There are 11 terpene synthase (TPS) and 45 potential polyketide synthase (PKS) genes annotated so far. However, numerous of the corresponding products and their physiological or ecological roles are still elusive. It has been demonstrated that polyketide-derived compounds such as DIF-1 and dictyodenes play a role in differentiation processes or spore dormancy, respectively. Although many structures of the amoebal sesqui- and diterpenes have been already identified by *in vitro* studies, their biological function remains to be investigated. Such volatile compounds are proposed to repel or attract insects or are involved in differentiation during the developmental cycle.

In this study, we aimed to explore the still elusive terpene products derived from TPS3 and investigate their function in an ecological context. Extrachromosomal expression in amoeba combined with a bioreactor-based fermentation led to the overproduction of the corresponding terpenes during vegetative growth, facilitating their structure elucidation and detailed physiological characterization. Subsequent NMR analysis identified the TPS3-derived metabolite as a novel sesquiterpene that was further investigated for biological activity. Choice assays of the volatile compound and the predatory nematode *Caenorhabditis elegans* showed a strong attraction, suggesting a rather ecological than physiological function. Hence, our research shed light on the intriguing mechanism behind the predator-prey interaction of social amoebae and nematodes. By isolating TPS3-derived products, we discovered a novel sesquiterpene that is responsible for the attraction of its predator *C. elegans*. This work deepens our understanding of molecular interspecies communication and how amoeba might domesticate their predators.

Pharmacological inhibition of ENT1 enhances the impact of specific dietary fats on energy metabolism gene expression

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Medium chain fatty acids are commonly consumed as part of diets for endurance sports, and as medicinal treatment in ketogenic diets where these diets regulate energy metabolism and elevate adenosine levels. However, a role for the equilibrative nucleoside transporter 1 (ENT1) activity responsible for adenosine movement across membranes in this process is not well understood. Here, we investigate ENT1 activity in controlling the effects of two dietary medium chain fatty acids (decanoic and octanoic acid), employing the tractable model system *Dictyostelium*. We show that genetic ablation of three ENT1 orthologues unexpectedly improves cell proliferation specifically following decanoic acid treatment. This effect is not caused by increased adenosine levels triggered by both fatty acids in the presence of ENT1 activity. Instead, we show that decanoic acid increases expression of energy related genes related to fatty metabolism in beta-oxidation, and pharmacological inhibition of ENT1 activity leads to an enhanced effect of decanoic acid on increasing expression of TCA cycle and oxidative phosphorylation components, with these effects validated using functional outcomes of mitochondria and lipid droplet quantification. Thus, our data show that ENT1 regulates the medium chain fatty acid-induced increase in cellular adenosine levels and the decanoic acid-induced expression of key metabolic enzymes in energy provision, to identify a key role of ENT1 proteins in metabolic effects of medium chain fatty acids.

Evolution of microRNAs in Amoebozoa and implications for the origin of multicellularity

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MicroRNAs (miRNAs) are important and ubiquitous regulators of gene expression in both plants and animals. They are thought to have evolved convergently in these lineages and hypothesized to have played a role in the evolution of multicellularity. In line with this hypothesis, miRNAs have so far only been described in few unicellular eukaryotes. Here, we investigate the presence and evolution of miRNAs in Amoebozoa. This large monophyletic group includes strictly unicellular amoebae but also social amoebae that go through aggregative multicellular development. We focused our investigation on species belonging to *Acanthamoeba*, *Physarum* and dictyostelid taxonomic groups, representing a range of unicellular and multicellular lifestyles. miRNAs that adhere to both the stringent plant and animal miRNA criteria were identified in all examined amoebae, expanding greatly the total number of protists harboring miRNAs. We found conserved miRNAs between closely related species, but the majority of species feature only unique miRNAs. This shows rapid gain and/or loss of miRNAs in Amoebozoa, further illustrated by a detailed comparison between two evolutionary closely related dictyostelids, i.e. *D. discoideum* and *D. firmibasis*. Additionally, loss of miRNAs in the *D. discoideum* *drnB* mutant did not seem to significantly affect multicellular development and, hence, demonstrates that the presence of miRNAs does not appear to be a strict requirement for the transition from uni- to multicellular life.

In addition, I will briefly go through a separate study where we used a multi-omics approach to study mRNA-protein correlation during early development of *D. discoideum*. The data is available in an interactive web app for ease of use: <https://edelbroek2024.serve.scilifelab.se>.

Unlocking CRISPR-Cas9 editing for widely diverse Dictyostelid species

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Dictyostelids are a species-rich clade of cellular slime molds that are widely found in soils and have been studied for over a century. Most research focuses on *Dictyostelium discoideum*, which – due to its ease of culturing and genetic tractability – has been a leading model species for molecular discoveries in the fields of cell biology, developmental biology and microbiology. Over decades, genome editing methods in *D. discoideum* have steadily improved but remained limited in scope, effective in only a few strains and species. Here, we introduce a CRISPR-Cas9 editing protocol that is cloning-free, selection-free, highly-efficient, and effective across *Dictyostelid* species. After optimizing our protocol in *D. discoideum*, using homology-directed repair, we obtained knock-out efficiencies of ~80% and knock-in efficiencies of ~30% without selective marker. Efficiencies depend on repair template concentrations, insertion sizes, homology arms and target sites. Since our protocol is selection-free, we can isolate mutants as soon as one day post-transfection, vastly expediting the generation of knock-outs, fusion proteins and expression reporters. Our protocol also makes it possible to generate several knock-in mutations simultaneously in the same cells, simplifying the construction of complex combinatorial reporter strains. Boosted by single-cell sorting and fluorescent microscopy, we could readily apply our CRISPR-Cas9 editing protocol to phylogenetically distant *Dictyostelid* species, which diverged hundreds of millions of years ago and have never been genome edited before. Our protocol therefore opens the door to performing broad-scale genetic interrogations across *Dictyostelids*.

From Cytoplasm to Nucleus: New Insights into DIRS-1 Replication Cycle

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Transposable elements (TEs) are parasitic DNA sequences capable of moving and amplifying within host genomes. A major class of TEs, retroelements, replicate via an RNA intermediate through a copy-and-paste mechanism. The *Dictyostelium Intermediate Repeat Sequence 1* (DIRS-1) is a founding member of a distinct family of retroelements that uniquely employs a tyrosine recombinase (YR) for integration, rather than the conventional integrase (INT) used by other retroelements. In *Dictyostelium discoideum*, DIRS-1 mobility is tightly regulated by RNA interference (RNAi), primarily through the combined action of the Argonaute protein AgnA and the RNA-dependent RNA polymerase RrpC. Deletion of either *agnA* or *rrpC* leads to mobilization of DIRS-1, providing a powerful model to dissect its replication mechanism. Notably, the cytoplasmic processing steps and nuclear import of DIRS-1 cDNA remain poorly understood. To investigate these early steps, we overexpressed a YR-GFP fusion protein in an *agnA*– background and tracked the intracellular dynamics of the putative pre-integration complex (PIC). Our preliminary observations indicate that prior to nuclear translocation, the PIC is directed to the microtubule-organizing center (MTOC) via the microtubule network. Remarkably, the DIRS-1 cDNA intermediate—a molecular hallmark of active retrotransposition—also accumulates at the MTOC in this mutant background. These findings suggest that the MTOC may serve as a key intermediate site in the cytoplasmic phase of DIRS-1 replication and raise new questions about host interactions and trafficking mechanisms unique to this retroelement family.

How intracellular actin waves mediate topological guidance and curvature sensing

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The highly persistent motion of *D. discoideum*, reminiscent to that of keratocytes, is intrinsically linked to the presence of actin waves, which are traveling wave patterns that propagate along the cortex. Here, we study the migration of single cells on micropillars of different curvatures, and we investigate the interplay of topological guidance and curvature sensing. We show that actin waves act as stabilising structures along the pillar edge, resulting in a persistent curvature-guided movement. The motility mode of the cells following the edge is dependent on the curvature, as different pillar diameters are associated to either single- or multiple- waves tracking the edge. We find that the preferential migration along the edge is inhibited when cells are unable to produce actin waves, displaying instead the typical amoeboid movement. Our results suggest that the topological guidance via actin wave formation is therefore critical for the edge-tracking migration.

How do you Make a Macropinocytic Cup?

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Macropinocytosis, the bulk uptake of extracellular fluid, is used by a range of cells to acquire nutrients and information from the surrounding environment. How cells organise the large deformations of the plasma membrane required to engulf a droplet of extracellular fluid is however unclear. This is an important question as this pathway becomes a significant source of nutrients during cancer progression. Most of what we know about the organisation of the actin cytoskeleton during macropinocytosis comes from work in *Dictyostelium discoideum* where domains of the plasma membrane marked by the phosphoinositide $\text{Pi}(3,4,5)\text{P}_3$ act as a template for the formation of a macropinocytic structure. These domains are ringed by the Arp2/3 complex which generates a branched actin network which can push against the plasma membrane, leading to the formation of a stereotypical macropinocytic cup. How the Arp2/3 complex is organised in a ring around the patch of $\text{Pi}(3,4,5)\text{P}_3$ but excluded from the centre is however unclear, especially as Rac GTPases, canonical activators of the complex, are active throughout the domain. Here we test several hypotheses to explain ring formation in both *Dictyostelium* and mammalian cells and ultimately propose an unexpected role for the contractile actin cortex outside the $\text{Pi}(3,4,5)\text{P}_3$ domain in localising the Arp2/3 complex. We then consider the implications of this model beyond macropinocytosis in *Dictyostelium* such as in phagocytosis, cell migration and the formation of ventral actin waves, in both *Dictyostelium* and mammalian cells.

Intersystem Integration through Contractility Kits

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Biological functions are often studied in isolated systems, yet evidence suggests deeper interconnectivity. In *Dictyostelium discoideum*, pre-assembled cytoplasmic contractility kits (CKs)—complexes of cytoskeletal proteins—highlight a responsive and dynamic cytoskeletal role. These CKs accumulate at the cortex under mechanical stress, suggesting a broader function beyond structure and motility.

Recent work from our lab identifies two proteins—RNA-binding protein 1 (RNP1A) and methylmalonate-semialdehyde dehydrogenase (mmsdh)—as potential mediators of crosstalk between the cytoskeleton and other cellular systems. RNP1A contributes to cell mechanics and mechanoresponsiveness, binds mRNAs encoding macropinocytosis-related proteins, and alters macropinocytosis when perturbed. Proteomics show RNP1A interacts with key CK components, including cortexillin I and IQGAP1.

Mmsdh, a metabolic enzyme involved in amino acid catabolism, also influences cortical dynamics, cell behavior, and morphology. Notably, its overexpression rescues cytokinesis defects caused by phosphomimetic myosin II mutations. Proteomic data indicate mmsdh interacts with cortexillin I and myosin II, linking it to CK function. Conversely, cytoskeletal mutants present metabolic phenotypes associated with mitochondrial quality and ATP-producing pathways. Transcriptomic analysis further reveals a rewiring of metabolic pathways, indicating a broader shift in the cell's metabolic profile associated with cytoskeletal perturbation.

Together, these results support two working models: RNP1A coordinates macropinocytosis and cytoskeletal remodeling via CKs, while mmsdh integrates metabolic and cytoskeletal functions. These findings uncover new layers of intersystem communication embedded in cytoskeletal networks.

Multicellular development in complex environments

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Cells of all kinds work together in multicellular behaviors ranging from collective migration to development. Simple laboratory assays have revealed a number of different behaviors cells engage in and how they're coordinated. However, natural environments are more spatially complex than most laboratory assays in ways that change both how cells can coordinate with one another and what behaviors they might need to perform. To understand how the complexity of natural environments shapes multicellular coordination and behaviors, we focus on how the soil-dwelling microbe *Dictyostelium* communicates and behaves in environments with many different size particles that better mimic natural conditions. These assays reveal new behaviors cells engage in to move around their environment and cooperate in complex spaces, as well as remodel their environment. Our findings highlight the importance of interactions between environments and cells, showing how naturalistic environments don't just constrain, but can enable new forms of multicellular cooperation.

Impact of simulated heatwaves on multicellularity in *Dictyostelium discoideum*: A proteomics perspective

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The effects of heatwaves at organism and population levels have been widely investigated; however, little is known about how they affect the development of cell populations and the fitness of the resulting organism. Disruptions caused by heatwaves are especially critical during early developmental stages in organisms lacking parental developmental protection or care. *D. discoideum*, a soil microbe with a life cycle that transitions between single-cell and multicellular stages, thrives optimally at 22 °C and elevated temperatures impair (27 °C) or completely arrest (30 °C) growth, development, and spore yield. Using a simulated heatwave model (exposure to 27 °C for 3 days) we observed a significantly downregulated the expression of cyclic AMP-dependent early developmental genes (*carA*, *acaA*, *pkaR*, *gtaC*, *tgrC1*, and *csaA*), as well as prespore markers (*cotB* and *spiA*), while the prestalk marker *ecmA* was comparatively less affected. When mixed with heat-stressed cells, reporter cells grown at 22 °C preferentially occupied the spore mass of the fruiting body and chimera assays of wild-type and reporter cells grown at different temperatures confirmed decreased fitness of heat-stressed cells. To investigate in depth the changes elicited by a simulated heatwave we performed differential proteomics analysis and found that downregulated proteins were primarily associated with chemotaxis and sporulation, whereas upregulated proteins were associated with metabolic processes. No major changes were observed in proteins related to microtubule or contractile vacuole organization. Strong upregulation of retrotransposable elements under heat stress suggests a potential link to genome instability, which may contribute to impaired development and reduced fitness. We conclude that exposure of unprotected organisms at the single cell stage to a single heatwave can negatively impact their ability to cope with environmental extremes.

***D. discoideum* amoebae fed on poor prey proliferate less but do not lose in the competition to become spores rather than stalk cells**

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Even generalist consumers in heterogeneous environments find some prey more edible than others. Those predators, forced by circumstance to eat mostly suboptimal foods, may be weaker than others. This may have consequences in *Dictyostelium discoideum* fruiting body formation where stronger amoebae are more likely to become spores. We measured growth and spore formation of *D. discoideum* grown with three different prey types that vary in quality. We predicted that those feeding on poorer prey would be more likely to end up in the stalk, not spore. The three food bacteria were *Klebsiella pneumoniae*, a bacterium commonly used for *D. discoideum* cultures, *Pseudomonas* 20P.3.2.4, an unpalatable bacterium isolated from a *D. discoideum* fruiting body collected from deer feces at Mountain Lake Biological Station, and, third, a mutant of the latter with inserts in the *gacA* and *cyoE* genes which made it more edible. We grow amoebae on different prey to assess whether growth on poor food reduces their competitiveness. We found that the amoeba on unpalatable *Pseudomonas* produces fewer spores, has a decreased proliferation rate, which is also positively related to the initial *D. discoideum* population density (Allee effect). Spore size and viability remained unaffected, but amoebae hatched from spores formed after consuming unpalatable prey were slower to proliferate when subsequently exposed to palatable bacteria. In chimeric fruiting bodies, feeding history alone did not significantly alter spore contribution unless amoebae were exposed to the additional stress of the translational inhibitor G418, which harmed cells that were previously fed unpalatable *Pseudomonas* more than the others. Our findings suggest that although slow growth and reduced spore yield impose fitness costs, *D. discoideum* does not show immediate social disadvantages. However, negative effects observed after spores are transferred to palatable bacteria reveal the cost of generalist predation and challenge the expectation that sporulation resets all epigenetic effects from prior feeding.

Poster

First author alphabetical order

The use of the antibiotic nourseothricin as retrotransposition marker

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Retrotransposons are mobile genetic elements that increase their copy number in the genome through a "copy-and-paste" mechanism involving reverse transcription. In *Dictyostelium discoideum*, retroelements make up approximately 10% of the genome. Some of these retroelements are regulated by the RNA interference (RNAi) machinery.

Skipper is a long terminal repeat (LTR) retroelement that is enriched in centromeric regions. Traceable versions of Skipper were generated using the *mnatl* marker gene to monitor retrotransposition. A nourseothricin *N*-acetyl transferase gene (*nat*) with an intron in reverse orientation (*natl*) was cloned in minus strand direction (*mnatl*) into a full-length Skipper element. The *mnatl* gene is initially inactive due to the presence of the intron (master element); when spliced out during Skipper^{nat} transcription, the functional *mnat* gene is restored and confers resistance to nourseothricin after productive retrotransposition (copy element).

To investigate the retrotransposition competency of Skipper^{nat}, two vectors were constructed: a pDM304 containing both Skipper^{nat} and a G418 resistance gene, and another plasmid containing only Skipper^{nat}. These vectors were transformed into several *Dictyostelium* RNAi mutants, including *agnA*-, *drnB*-, and *rrpC*- and the AX2 wildtype. The first vector was introduced with or without the pISAR plasmid, which carries a G418 resistance gene, while the second vector was co-transformed with pISAR. G418 selection was used to confirm successful transformation, followed by a second selection using nourseothricin to enrich cells with Skipper^{nat} copy elements.

In plates where cells survived the second selection, the intron had been spliced out, indicating that Skipper^{nat} was functionally expressed and retrotransposition competent. PCR products of the *mnat* gene showed double bands for master and copy elements for the *agnA*-, *drnB*-, and *rrpC*- strains. Surprisingly, also a wildtype strain AX2 showed a double band, but only if the single plasmid encoding Skipper^{nat} and the G418 resistance gene was used together with the pISAR plasmid.

The plasmid encoding Skipper^{nat} only was also introduced into other RNAi mutants such as *rbdB*-, *agnB*-, and *agnA/B*- strains and co-transformed with either the pDM304 plasmid and or the pISAR plasmid. In the *rbdB*-, *agnB*-, and *agnA/B*- strains, only non-functional *mnatl* was observed when using pDM304. However, when using the pISAR plasmid, both master and copy element were detected in strains surviving the second selection.

The data suggest that the *mnatl* gene is, in principle, suitable for tracking the amplification of retrotransposons in *D. discoideum* cells. However, the use of the *mnatl* gene yielded some conflicting results regarding the roles of individual components of the RNAi machinery in the silencing of Skipper mobility, warranting further investigation.

Heterologous production of cannabinoid precursors in amoeba by genomic integration of biosynthetic plant genes

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Communication via small molecules is vital for the life cycle and niche adaptation of *Dictyostelium discoideum*. The identity and biosynthesis of such molecules has been the focus of researchers over the last decades. However, the potential of *D. discoideum* to produce such complex molecules, which are also of pharmaceutical interest, like polyketides and terpenes, has rarely been investigated. We have previously exploited the biosynthetic capacity of *D. discoideum* to express type III polyketide synthases (PKSs) from plants and produce aromatic specialized metabolites. Additionally, we engineered a hybrid PKS based on the *D. discoideum* native PKS Steely B (StlB/PKS37), which was fused to the type III PKS olivetol synthase (OLS) from *Cannabis sativa*. By simultaneous expression of this PKS and the olivetolic acid cyclase (OAC) from *C. sativa*, we demonstrated *in vivo* biosynthesis of olivetolic acid (OA) in amoeba. OA itself is a vital intermediate in the biosynthetic pathway of cannabinoids, which have garnered interest for pharmaceutical applications.

Based on this new synthesis pathway of OA, we aim to establish a production platform for cannabinoid-related compounds in *D. discoideum*. By stable genomic integration of the genes encoding the PKS37/OLS hybrid and the OAC, we achieved an increase of the overall OA production from the initially reported 4.5 µg/L to 375 µg/L. To complete the cannabinoid biosynthesis in amoeba, we finally engineered and expressed genes encoding the prenyltransferase PT4 and tetrahydrocannabinolic acid synthase THCAS from *C. sativa* and proved *in vitro* functionality of the downstream biosynthetic pathway.

Exploring the function of the protein-sorting receptor sortilin in *Dictyostelium discoideum*

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Intracellular protein sorting is essential for supplying resources needed by the organelles of a cell. For example, lysosomes require enzymes that break down bulk material like food and defective cellular components. Enzymes are trafficked to lysosomes by sorting receptors that are embedded in the membrane of the Golgi apparatus. Sorting receptors are well characterized in mammals but have been elusive in *Dictyostelium discoideum*. For example, a homolog of the human sorting receptor sortilin (SORT1) was only recently identified in *D. discoideum* (Sort1). We obtained a *sort1* knockout mutant from the GWDI bank and have begun to characterize its cellular role during the *D. discoideum* life cycle. Cells lacking *sort1* display increased plaque growth when placed on SM/2 agar containing a bacterial lawn. Interestingly, this phenotype is reversed in HL5 liquid culture where cells lacking *sort1* fail to proliferate and become significantly less viable. In support of these observations, western blots revealed reduced intracellular amounts of the quorum-sensing proteins autocrine proliferation repressor A (AprA) and counting factor-associated protein D (CfaD), as well as the density-sensing factor countin (CtnA). Intriguingly, these reduced protein amounts are only observed in cells harvested from cells germinated in HL5, and not in bacterially-fed cells harvested from SM/2 agar. In addition, the intracellular amount of the lysosomal enzyme cathepsin D (CtsD) is reduced only in cells germinated in HL5. These results suggest *D. discoideum* uses Sort1 to regulate its growth, especially in environments that require macropinocytosis. Furthermore, this is the first evidence in *D. discoideum* linking Sort1 to lysosomal enzymes, which further supports its identification as a SORT1 homolog. Lastly, we have generated a custom antibody against Sort1, which we used to demonstrate that Sort1 is glycosylated. Together, this research lays the foundation for establishing *D. discoideum* as a model organism for studying sortilin-mediated protein trafficking.

Biohybrid Microtransport: Integrating Cellular Motility with Microparticle Dynamics

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Cell-driven micro-transport — the movement of micron-sized particles by motile cells — is a key example of composite active matter. *Dictyostelium discoideum*, a well-established model organism in microbiology, has been shown to transport passive spherical microparticles through chemotaxis and collective behavior, establishing mechanical contact. In recent work, we estimated the transport forces under different environmental conditions and developed an active particle model that provides estimates for the long-term transport coefficients.

In this study, we present new insights into how *D. discoideum* interacts with particles that are not of spherical shape — a regime largely unexplored. Time-lapse imaging shows that cells engage with these particles in highly exploratory, dynamic ways, reminiscent of infants discovering new objects. These interactions lead to active repositioning and complex motion patterns not captured by existing models developed for larger, spherical cargo. It is our overall long-term goal to investigate how particle shape affects displacement statistics, integrating physical transport models with experimental observations. At the poster, we will discuss first descriptive findings in the light of current models of active microtransport.

Intersystem Integration through Contractility Kits

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Biological functions are often studied in isolated systems, yet evidence suggests deeper interconnectivity. In *Dictyostelium discoideum*, pre-assembled cytoplasmic contractility kits (CKs)—complexes of cytoskeletal proteins—highlight a responsive and dynamic cytoskeletal role. These CKs accumulate at the cortex under mechanical stress, suggesting a broader function beyond structure and motility.

Recent work from our lab identifies two proteins—RNA-binding protein 1 (RNP1A) and methylmalonate-semialdehyde dehydrogenase (mmsdh)—as potential mediators of crosstalk between the cytoskeleton and other cellular systems. RNP1A contributes to cell mechanics and mechanoresponsiveness, binds mRNAs encoding macropinocytosis-related proteins, and alters macropinocytosis when perturbed. Proteomics show RNP1A interacts with key CK components, including cortexillin I and IQGAP1.

Mmsdh, a metabolic enzyme involved in amino acid catabolism, also influences cortical dynamics, cell behavior, and morphology. Notably, its overexpression rescues cytokinesis defects caused by phosphomimetic myosin II mutations. Proteomic data indicate mmsdh interacts with cortexillin I and myosin II, linking it to CK function. Conversely, cytoskeletal mutants present metabolic phenotypes associated with mitochondrial quality and ATP-producing pathways. Transcriptomic analysis further reveals a rewiring of metabolic pathways, indicating a broader shift in the cell's metabolic profile associated with cytoskeletal perturbation.

Together, these results support two working models: RNP1A coordinates macropinocytosis and cytoskeletal remodeling via CKs, while mmsdh integrates metabolic and cytoskeletal functions. These findings uncover new layers of intersystem communication embedded in cytoskeletal networks.

Exploring a potential role for *Dictyostelium* Stat proteins in phagocytosis and chemotaxis

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The *Dictyostelium* Stat (signal transducers and activators of transcription) proteins StatA, StatB, and StatC are known to regulate gene expression during development and, in StatC's case, in response to osmotic stress. Here we made use of *statA*⁻, *statB*⁻, and *statC*⁻ null mutants to explore a possible role for Stat proteins in phagocytosis (as assessed via plaque formation on a *Klebsiella* lawn) and chemotaxis to folate. *statB*⁻ mutants did not differ from wild-type (Ax2) cells, suggesting that StatB is not required for these processes. *statA*⁻ mutants formed unusually large plaques on bacterial lawns. And, while *statA*⁻ cells displayed directed movement toward folate, both the relative area of migration and length of migration toward the folate source exceeded that of Ax2. In contrast, consistent with prior reports, *statA*⁻ cells starved for 4 hrs were less effective than Ax2 in responding to cAMP. The starved cells also demonstrated random outward movement in controls in which KK2 replaced cAMP. Additional analysis would be required to determine whether the mutant's random movement following starvation, seeming hypersensitivity to folate, and expanded plaque size reflect a common inhibitory effect of StatA on cell movement. Interestingly, StatA has recently been identified as one in a set of proteins phosphorylated by Erk2 in response to both folate and cAMP stimulation. Similar to *far1*⁻ mutants (which lack fAR1, the receptor for folate), *statC*⁻ cells formed very small plaques on bacterial lawns and were defective in chemotaxis towards folate, while retaining the ability to respond to a cAMP gradient. We are currently exploring whether folate can induce nuclear localization of StatC in a fAR1-dependent manner, as would be predicted if folate, like DIF-1 and osmotic stress, serves as an activator of StatC. Taken together, our findings reveal previously unknown roles for StatA and StatC, particularly in folate-dependent processes in vegetative-stage cells.

Exploring the volatile bouquet of social amoeba

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The discovery of novel natural products is essential for modern pharmaceuticals and the development of new therapeutic agents. [JK1] The social amoeba *Dictyostelium discoideum* presents a unique opportunity to explore bioactive compounds due to its complex lifecycle and its role as a model organism in biological research and natural product discovery. *D. discoideum* harbors numerous biosynthetic genes, including 11 terpene synthase (TPS) encoding genes that predominantly produce sesquiterpenes and diterpenes. Most of them are expressed during later developmental stages and might play a potential role in cell signaling or defence. The TPS3 encoding gene, for instance, is highly expressed during slug and fruiting body formation and is subject to strict transcriptional regulation, suggesting important biological functions of its still unknown derived products.

To shed light on terpene production in amoeba, we first started with the constitutive overexpression of the TPS5 gene as a *proof-of-principle* and demonstrated the successful *in vivo* production of the already described diterpene β -pinacene. We further overexpressed the TPS3 encoding gene in *D. discoideum* to chemically and functionally characterize its elusive metabolites. GC/MS analysis revealed several volatile TPS3-derived products, mainly sesquiterpenes, with two of them being unidentified compounds. Furthermore, volatile profiling during the development showed the same products as found in the vegetative stage, indicating a final product formation of TPS3 without additional modifications during the developmental cycle. For compound isolation, the TPS3 production was scaled up in a 25 L bioreactor, which enabled NMR-guided structure elucidation, revealing the major compound as a novel sesquiterpene. Exposure assays of the purified TPS3 products with the predatory nematode *Caenorhabditis elegans* showed a strong attractant effect, which suggests a potential role in interspecies communication.

Stripes of cell group deposition at the feeding front

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Dictyostelium cells shed clumps of cells as they migrate towards their bacterial food source. We have identified cell strains and conditions that cause the clumps to be shed in regularly spaced lines from the feeding front. Under different boundary conditions, clump shedding switches to a pattern with non-parallel lines reminiscent of the seeds in a sunflower. I will discuss our fluid dynamic models explaining this non-Turing pattern formation mechanism.

Temporal Changes in Nuclear Envelope Permeability during Semi-Closed Mitosis in *Dictyostelium* Amoebae

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Dictyostelium discoideum performs a semi-closed mitosis in which the nuclear membranes remain intact but its nuclear envelope (NE) becomes permeabilized at the onset by partial disassembly of the nuclear pore complexes (NPCs) to allow tubulin and spindle assembly factors get access to the nuclear interior. The dynamic processes also involve the insertion of the mitotic centrosome into the NE and a further fenestration in late mitosis at the abscission sites of the two daughter nuclei.

In addition to their function in NPCs, we have investigated the potential involvement of nucleoporins in the maintenance and restoration of nuclear envelope permeability.

The transmembrane Nup210, is present at the dividing nuclei throughout mitosis and is concentrated at the fenestrae harboring the mitotic centrosomes together with the LINC complex protein Sun1 that forms a ring-like structure around the inserted centrosomes. Nup210 lasts there until late telophase when the centrosome is positioned towards the cytosolic side of the NE again. Sun1 is required for anchoring the centrosome to the nucleus in interphase and might contribute together with Nup210 in holding the centrosome in place during mitosis. A potential interaction of both proteins was found in a BioID-assay.

The inner ring Nup93, dissociates from the NE during prophase and reappears in late anaphase. During late mitosis, it is enriched at the abscission site around the still existing central spindle together with the HeH/LEM-family transmembrane protein Src1, an interactor that was identified by BioID and mass spectrometry.

In yeast, extrusion of the centrosome and sealing of the NE at the end of mitosis is initiated by the interaction of Heh1 and the ESCRT-III like protein CHMP7. This complex recruits further ESCRT-III subunits that polymerize and form a helix. The AAA-ATPase VPS4 drives filament remodeling and membrane constriction to close the holes. In *Dictyostelium* all components localize to the sealing sites that makes it likely that it uses the same pathway for NE closure.

A Resource Platform from NBRP Nenkin for the *Dictyostelium* Community

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The National Bioresource Project for Cellular Slime Molds (NBRP Nenkin) has served as a key bioresource center in Japan since 2007, with the aim of supporting the global *Dictyostelium* research community. Since 2022, it has been hosted by the University of Tsukuba. NBRP Nenkin collects, preserves, and distributes cellular slime mold strains and plasmids, particularly those derived from Japanese laboratories engaged in diverse biological research. To date, we have archived over 2,100 strains and 1,000 plasmids. This collection includes all-in-one CRISPR/Cas9 vectors for gene editing applications, such as knockouts, inducible knockouts, knockdowns, knock-ins, point mutations, and deletions, which are in high demand among researchers worldwide. We also provide Flamindo2, a highly sensitive fluorescent biosensor that allows real-time monitoring of cytosolic cAMP concentrations ranging from ~0.3 to 12 μ M during early developmental stages and other useful various fluorescent marker vectors. Additionally, our collection features field-isolated strains that are being utilized in pharmacological screenings, including the identification of potential drug lead compounds. All resources are publicly accessible via our website (<http://nenkin.nbrp.jp>), and we actively welcome deposits of new resources from researchers around the world. NBRP Nenkin aims to complement existing platforms such as DictyBase by offering unique resources and fostering open exchange within the *Dictyostelium* community. We invite researchers to explore, utilize, and contribute to this expanding platform.

The proteome of *Dictyostelium discoideum* across its entire developmental cycle

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Dictyostelium discoideum is a widely used eukaryotic model organism in evolutionary and cell biology due to its ability to transition from a single cell to a multicellular form. While gene expression information across the developmental cycle is widely available at transcript level, only early-stage data exists at proteome level. To address this knowledge gap, we performed label free quantitative proteomics on a time series of *D. discoideum* developing synchronously on nitrocellulose filters. Samples were collected and analyzed at the vegetative, aggregation, mound, culmination and fruiting body stages. We identified a total of 4,503 proteins, of which 1,848 (41%) were present across all stages. Gene Ontology analysis showed key functions associated with each stage: tRNA aminoacylation and translation in the vegetative stage, ribosomal biogenesis in aggregation, actin filament capping and Ras signaling in the mound, sorocarp development in the culmination and extracellular matrix organization in fruiting body stages. Adjacent stage transitions were analysed using pairwise comparisons. This analysis revealed that the largest transitions occurred between culmination and fruiting body and between fruiting body and vegetative stage, with 46% and 67% of differentially abundant proteins, respectively. To investigate the behaviour of the proteins across the whole developmental cycle we used k-means clustering followed by functional mapping of 9 clusters, each displaying a distinct pattern of regulation. This study provides, for the first time, a complete developmental proteomic time series for *D. discoideum*, revealing protein level changes that contribute to multicellularity, cellular differentiation and morphogenesis.

Impact of simulated heatwaves on multicellularity in *Dictyostelium discoideum*: A proteomics perspective

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The effects of heatwaves at organism and population levels have been widely investigated; however, little is known about how they affect the development of cell populations and the fitness of the resulting organism. Disruptions caused by heatwaves are especially critical during early developmental stages in organisms lacking parental developmental protection or care. *D. discoideum*, a soil microbe with a life cycle that transitions between single-cell and multicellular stages, thrives optimally at 22 °C and elevated temperatures impair (27 °C) or completely arrest (30 °C) growth, development, and spore yield. Using a simulated heatwave model (exposure to 27 °C for 3 days) we observed a significantly downregulated the expression of cyclic AMP-dependent early developmental genes (*carA*, *acaA*, *pkaR*, *gtaC*, *tgrC1*, and *csaA*), as well as prespore markers (*cotB* and *spiA*), while the prestalk marker *ecmA* was comparatively less affected. When mixed with heat-stressed cells, reporter cells grown at 22 °C preferentially occupied the spore mass of the fruiting body and chimera assays of wild-type and reporter cells grown at different temperatures confirmed decreased fitness of heat-stressed cells. To investigate in depth the changes elicited by a simulated heatwave we performed differential proteomics analysis and found that downregulated proteins were primarily associated with chemotaxis and sporulation, whereas upregulated proteins were associated with metabolic processes. No major changes were observed in proteins related to microtubule or contractile vacuole organization. Strong upregulation of retrotransposable elements under heat stress suggests a potential link to genome instability, which may contribute to impaired development and reduced fitness. We conclude that exposure of unprotected organisms at the single cell stage to a single heatwave can negatively impact their ability to cope with environmental extremes. and reporter cells grown at different temperatures confirmed decreased fitness of heat-stressed cells. To investigate in depth the changes elicited by a simulated heatwave we performed differential proteomics analysis and found that downregulated proteins were primarily associated with chemotaxis and sporulation, whereas upregulated proteins were associated with metabolic processes. No major changes were observed in proteins related to microtubule or contractile vacuole organization. Strong upregulation of retrotransposable elements under heat stress suggests a potential link to genome instability, which may contribute to impaired development and reduced fitness. We conclude that exposure of unprotected organisms at the single cell stage to a single heatwave can negatively impact their ability to cope with environmental extremes.

Talin force coupling underlies eukaryotic cell adhesion

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Integrin-mediated cell adhesion and mechanotransduction are considered key innovations in animal evolution. Here, we show that these processes represent a specialization of an evolutionarily conserved force coupling mechanism that originated in unicellular eukaryotes and is mediated by the actin-binding protein talin. In contrast to heterodimeric integrin receptors, talin is widely distributed in unicellular organisms, including amoebae. By comparing the molecular mechanics of talin-A from *Dictyostelium discoideum* with those of mammalian talin-1, we uncover a conserved role for talin in transmitting pN-scale forces into cells, even in unicellular organisms lacking integrin receptors. Our data demonstrate that not only the association of the talin-A rod with the actin cytoskeleton, but also its engagement with transmembrane receptors – such as SibA and SadA – are both required for efficient force coupling and are indispensable in the vegetative state. Our most recent experiments reveal that talin-B is found in intercellular adhesion in developmental stages, suggesting an unexpected role for talin in the evolution of cell-cell adhesion.

***Dictyostelium discoideum* as a Model System for Probiotic Screening: Dissecting Host–Bacteria Interactions**

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Monoclonal antibodies are widely used for the treatment of patients affected by follicular lymphoma (FL) and multiple myeloma (MM). However, prolonged exposure to monoclonal antibodies can impair immune function, increasing the risk of infections. For this reason, the use of probiotics is gaining importance as a strategy to counteract pathogenic colonization. Among them, *Streptococcus salivarius*, a natural inhabitant of the respiratory tract microbiota, has been previously identified by our group as a promising oral probiotic. In this project, we propose the use of the social amoeba *Dictyostelium discoideum* (hereafter *Dictyostelium*) as a simple and cost-effective model to study host–bacteria interactions. Owing to its innate ability to feed on bacteria via phagocytosis, the probiotic effects of *S. salivarius*, as well as other bacterial strains, were assessed through *Dictyostelium* growth, vitality and phagocytosis assays, along with gene expression and metabolomic analyses. Our results show that both *S. salivarius* and the oral pathogen *Streptococcus mutans* reduced *Dictyostelium* growth compared to *E. coli* (standard control), but only *S. salivarius* enhanced cellular vitality, suggesting a stimulatory probiotic effect. While phagocytosis was impaired by *S. mutans*, both strains modulated genes involved in oxidative stress and lysosomal function. Moreover, metabolomic profiling revealed a more active and balanced metabolic state in cells treated with the probiotic, whereas *S. mutans* induced a more stressed and energy-depleted profile. To further validate these observations, chemotaxis and RNA-Seq analyses are currently underway. Overall, these findings support the use of *Dictyostelium* as a model for probiotic screening and highlight the contrasting effects of commensal versus pathogenic bacteria on host-like responses. Ultimately, *Dictyostelium* stands out as an accessible and versatile system to investigate how probiotics influence fundamental cellular processes.

Establishing *Dictyostelium* as a model system to investigate *M. abscessus* virulence mutations

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Mycobacterium abscessus is a rapidly growing, non-tuberculous mycobacterium (NTM) that has emerged as an important opportunistic pathogen, particularly in individuals with underlying lung conditions such as cystic fibrosis. It is highly resistant to many antibiotics, making infections difficult to treat and a growing concern in clinical settings. *Dictyostelium discoideum* is an established model system to study mycobacterial virulence mechanisms.

Its genetic tractability, phagocytic lifestyle, and conservation of key cellular pathways make it a valuable tool for investigating host-pathogen interactions and the basic mechanisms of cellular immune responses.

Using several independent experimental approaches, we can show that, in contrast to non-pathogenic *M. smegmatis*, both smooth and rough *M. abscessus* strains can survive within *Dictyostelium* cells. Infections of an autophagy mutant ($\Delta atg1$) suggest that autophagy may play a role in susceptibility to *M. abscessus* growth. We have a collection of *M. abscessus* isolates from longitudinal samples from cystic fibrosis patients available, which exhibit mutations selected within the lung during chronic infection. *Dictyostelium* will allow us to further investigate these mutants and characterize their fate within host cells.

Brakes in the Cup: Spatial Control of Actin Polymerization during Macropinocytosis

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Macropinocytosis is an evolutionarily conserved, non-selective endocytic pathway where cells use actin-driven membrane ruffles to engulf large volumes of extracellular fluid. *Dictyostelium discoideum*, which relies heavily on macropinocytosis for feeding, serves as a powerful model system for studying this process and the underlying regulation of membrane dynamics and actin assembly.

Macropinocytic cups form around phosphatidylinositol (3,4,5)-trisphosphate (PIP₃)/Ras/Rac signaling domains. At their edges, the Scar/WAVE–Arp2/3 pathway drives branched actin polymerization, but how this activity is spatially confined remains unclear. What stops actin from polymerizing throughout the entire domain?

To address this, I investigate two negative regulators of actin branching: CYRI, which sequesters Rac1 and inhibits the Scar/WAVE complex, and Arpin, which directly suppresses the Arp2/3 complex. Although both proteins have been studied in migration and phagocytosis, their roles in macropinocytosis remain largely unknown.

This work provides the first functional analysis of CYRI and Arpin in macropinocytosis, shedding light on how actin dynamics are spatially and temporally controlled to maintain the structure and efficiency of this fundamental cellular process.

Differential kinetics of self organizing branched actin structures associates with upstream signaling pathway in *Dictyostelium discoideum*

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Self-organizing cortical actin filaments gives living cells their shape, orchestrates their division, and allows them to move and interact in response to their mechanochemical environment. In response to most external cues, the cortical actin structure remodelling is initiated by Arp2/3 complex upon activation by nucleation promoting factors (NPFs) such as Wiskott-Aldrich syndrome protein (WASP), Suppressor of cyclic AMP receptor (SCAR) proteins. Within a cell, these NPFs are often spatio-temporally separated and are involved in different cellular processes. The important question raised in the field of branched actin dynamics is how the same Arp2/3 complex, activated by different NPFs, produces actin structures with different geometric and kinetics? *Dictyostelium discoideum* is a very unique system for understanding these branched actin structures. Normally, *Dictyostelium* SCAR and WASP are localized in different places and are involved in different cellular processes. But in the absence of SCAR, WASP can take over the role of SCAR in pseudopods. However, the functions that these structures enable depends not only on their mere existence, but also on many subtle aspects such as lifetime, dynamics, stoichiometry and forces that they support. And yet these aspects have not been explored quantitatively for the seemingly similar structures. In the present study we set out to explore these aspects of the transient branched actin structure formed in the cortical meshwork (cortical puncta). In *Dictyostelium*, these cortical puncta show the presence of F-actin, WASP and Arp2/3. Knockout studies revealed that punctas are bigger in size but with relatively similar lifetime in *scar*⁻ cells compared to wildtype. However in *wasp*⁻ cells, the puncta size remains the same but the puncta live for a longer time before they disappear. These preliminary results suggest the kinetics of branched actin structure formation differ depending upon upstream NPFs.

Optogenetic modulation of cAMP oscillations reveals frequency-selective cellular behaviours and transcription factor dynamics

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Oscillatory signalling dynamics play a crucial role in cellular decision-making during spatially and temporally regulated developmental processes. However, it remains poorly understood how their complex periodicity drives collective cell behaviours and gene expression. To address this issue, we established an optogenetic system combining a light-activated adenylate cyclase (mPAC) and a fluorescent cAMP biosensor (Pink Flamindo), enabling precise control and live imaging of cAMP dynamics in *Dictyostelium*. Employing this system, we revealed that 4-minute cAMP oscillations promoted more efficient cell aggregation relative to 6-minute oscillations. In contrast, modulation of oscillation frequency from 6 to 4 minutes suppressed the nuclear translocation dynamics of a cAMP-responsive transcription factor, GtaC, which governs developmental gene regulation. These findings suggest that while frequency-modulated cAMP signals promote cellular behaviours, the transcription factor acts as a low-pass filter to precisely control gene expression timing and ensure proper developmental progression.

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Travel:

By air travel

- Book your flight to Berlin Airport BER. Alternatively, you may fly to Frankfurt (Main) FRA and continue your travel by train (however approx. 5.5 hours train ride).

Unfortunately, the German railways have decided to carry out repair work on the line connecting BER airport and station Pirschheide. The resulting detour adds an hour to reach the hotel.

These are the most convenient connections for Sunday, August 3rd:

- **Bus RB22/Train RB22**; The single fare ticket "Berlin ABC" is valid for connections from Berlin to Potsdam (4.70 €).

BER airport	27 minutes past every hour	In front of airport
Ludwigfelde-Struveshof	+ 1 hour.27 min	
Ludwigfelde-Struveshof	+ 1 hour.35 min	Rail Track 2
Potsdam-Pirschheide	+ 1 hour.46 min	

- **Bus BER2 to Potsdam/Tram 91* to Pirschheide**; The single fare ticket "Berlin ABC" is valid for connections from Berlin to Potsdam (4.70 €) plus an additional charge (5 €) for bus BER2.

BER airport	9.08	10.36	11.36	13.10	14.10	15.40	17.10	Bus platform A2
Potsdam main (Hauptbahnhof)	10.08	11.36	12.36	14.10	15.10	16.40	18.10	
Potsdam main (Hauptbahnhof)	10.18	11.41	12.41	14.18	15.18	16.48	18.18	Bus platform 2
Potsdam-Pirschheide	10.27	11.48	11.48	14.38	16.38	17.13	18.38	

*Tram 91 stops also at station Luftschiffhafen, which is closer to Villa Carlshagen (Sunday evening welcome reception). If you miss Tram 91, you may take bus 631 in direction Werder (platform 8) or bus 580 in direction Brandenburg (platform 8). If using these two lines, alight at station Luftschiffhafen and either walk to Villa Carlshagen or to the Kongresshotel.

For the return:

After the weekend, the normal connection is in place again, leaving at Potsdam-Pirschheide at 14 minutes past every hour, reaching the BER airport in 22 minutes

- A taxi from the airport to the venue will cost at least 100 €. Please note that unlicensed drivers are approaching travellers within the airport building – please stay away! Official taxis wait outside the airport building.

By train

- Book your train journey to Potsdam Station (Hauptbahnhof). Connections can be found here: <https://int.bahn.de/en>

By car

- Use your GPS or the google maps route planner.

How to reach the hotel and the venue of the welcome reception (Sunday evening)

- In Potsdam station, use exit Friedrich-Engels-Strasse. Single fare tickets in Potsdam are 2.80 € and can be purchased from vending machines in buses and trams, which leave in front of the train station.
- Take tram 91 in direction Pirschheide (platform 2), bus 631 in direction Werder (platform 8) or bus 580 in direction Brandenburg (platform 8).
- In either case, alight at the stop Luftschiffhafen. At the tram stop you will find signs indicating the way to the hotel (Yellow route in map; 700 meters).
- The welcome reception on Sunday, August 3rd starts at 6pm at the nearby Villa Carlshagen, home of Health and Medical University (Address: Olympischer Weg 1). Depending on your arrival time, feel free to check in at the hotel first or to go directly to Villa Carlshagen from the bus and tram stop Luftschiffhafen (red route in map, 300 meters). It is an easy and pleasant walk between Villa Carlshagen and the Kongresshotel along lake Templin (purple route in map).



Maps:

Find a map of Potsdam here: Tickets "Berlin ABC" are valid in whole Potsdam and Berlin. 24 h tickets are available

<https://www.openstreetmap.org/search?query=Potsdam#map=12/52.4284/13.0275>

Public transportation:

VBB is a tariff joint region between the public transportation services of Berlin (BVG) and Potsdam (ViP). Berlin ABC Tickets are valid in whole Berlin and Potsdam, however the cheaper Potsdam ABC Tickets are valid in Potsdam only. Thus, with Berlin ABC you are on the safe side.

Use the free VBB-App "Bus&Bahn" (via Google or Apple app store) or the following website to find your connection and the required tickets.

<https://www.vbb.de/en/>

You can pay via the app if you have made a payment connection to Handy Ticket Deutschland within the app (you will need a credit card or a Paypal account). Otherwise you can pay cash at the ticket machines within the tram. Be prepared that you need coins or small banknotes. Debitcards may work, credit cards not. **Do not buy tickets *within* the tram *in advance* since they become validated automatically. Tickets bought at the ticket machines *outside* trams and busses (e.g. at train/bus/tram stations) are not yet validated and can be purchased in advance. They need to be validated at the ticket validator machines inside the trams and busses once you start your trip.** VBB tickets are valid also in regional trains (train no. starts with RE...) as long as you travel between Potsdam and Berlin.

Tram timetable from Pirschheide to the city center and back

https://www.swp-potsdam.de/content/verkehr/pdf_7/fahrplanwechsel_2025/tram_91.pdf